

The Selected $K3$ Surfaces of the Cooper s_7/s_{10} Families: Identification, the $K3 \times T^2$ Reading, and the Twisted-Weierstrass Route

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Stream 2 — Selection & Geometry (SocrateAI-Scientific-Agora-K3-DarkMatter)

2026-10-08 (preprint; not reviewed; continues the 2026-09-29 note)

What is not claimed

This is a mathematics note. It makes no physical claim: no dark-matter identification, no dark-energy statement, no observable. No point of either family maps to any measurement, the flux/tadpole route stays blocked until a threefold base is specified, and the physical reading of “ $K3 \times T^2$ ” stays **Tier C — blocked physics**. Nothing here ranks the s_7 family against the s_{10} family. No Kodaira fibre type is attached to any point of either family; we report discriminant-root orders and root lattices only.

Abstract

The 2026-09-29 note adopted, by an explicit ruling, a within-family selector on the $\rho = 20$ points of the Cooper s_7 and s_{10} Picard–Fuchs families: the point of minimal $|\text{disc } T|$. This note records what happened next. First, the selected points are identified as named surfaces. Through the Shioda–Inose correspondence between singular $K3$ surfaces and even positive-definite binary forms, and Takatsu’s tabulation, the s_7 pick ($T = A_2$, $\det T = 3$) is the unique singular $K3$ surface X_3 and the s_{10} pick ($\det T = 4$) is the unique X_4 . The pair is the one called “Vinberg’s two most algebraic $K3$ surfaces” in the Schütt–Shioda survey. Second, we state the one reading of “ $K3 \times T^2$ ” under which a selection is meaningful, the Shioda–Inose pairing with a pair of n -isogenous elliptic curves, and report the unscored consistency check between the level read from the lattice and the level read from the modular side. Third, we record the twisted-Weierstrass route. A degree question that had conflated two quantities is resolved, a two-divisor degree screen on candidate threefold bases is reported, and the Mordell–Weil section required of the M_7 -polarized member is computed. It has height 14 and $\bar{P} \cdot \bar{O} = 5$ generically and at $z = 1/27$, and $\bar{P} \cdot \bar{O} = 0$ at $z = -1$. The selected point $z = \infty$ has Mordell–Weil rank 0 and carries no such section. At $z = 1/27$ an explicit endomorphism of square -7 is written over $\mathbb{Q}$. Every table is generated from a certificate.

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1 Scope

Ledger rules of this programme bound everything below. The $\rho = 20$ cut and the discriminant-floor selector are naming conventions inside one family, adopted by explicit rulings; they do not compare the two families. The agreement between the binary-form side and the modular side of any statement here is forced by the Shioda–Inose structure (an elliptic point of the modular curve is a CM point) and is never offered as corroboration. The s_{10} lattice certificate is LIVE since 2026-09-29; that changes no value below and ranks nothing. The next step of the twisted-Weierstrass route that would involve a threefold (work package TW3) has not been opened, because no ruling specifies twist data over a threefold base.

2 The selection, in one table

Table 1 repeats the six $\rho = 20$ points of the 2026-09-29 note, regenerated from the current certificates. The lattice arithmetic of every row is kernel-checked in Lean 4 (Tier A); the identification of the orthogonal complement with the family’s transcendental lattice T is Tier B — derived, not measured (Dolgachev §7, Doran Thm. 5.13, Tier L — literature). The selector picks, in each family, the row of minimal $|D|$: $z = \infty$ in both.

Family	z	v	v^2	D	T (reduced)	Tier / sources
cooper_s7	-1	$(2, -4, 1)$	-2	-7	$[2, 1; 1, 4]$	Tier A, 2 source(s)
cooper_s7	$1/27$	$(1, -1, 0)$	-2	-28	$[2, 0; 0, 14]$	Tier A, 2 source(s)
cooper_s7	∞	$(14, -14, -5)$	-42	-3	$[2, 1; 1, 2]$	Tier A, 2 source(s)
cooper_s10	$-1/4$	$(2, -6, 1)$	-4	-20	$[4, 2; 2, 6]$	Tier A, 1 source(s)
cooper_s10	$1/16$	$(1, -1, 0)$	-2	-40	$[2, 0; 0, 20]$	Tier A, 1 source(s)
cooper_s10	∞	$(10, -10, -3)$	-20	-4	$[2, 0; 0, 2]$	Tier A, 2 source(s)

Table 1: The six $\rho = 20$ points. Compared with the 2026-09-29 build, the s_{10} rows no longer carry an “(ADVISORY)” tag; see section 7, item 1.

3 Which surfaces were selected

A singular $K3$ surface is one of Picard number 20. Shioda and Inose, in the form we read in Shioda’s survey and in Kumar–Kuwata (Thm. 8.1), give a bijection between singular $K3$ surfaces up to isomorphism and even positive-definite binary lattices up to $\mathrm{SL}_2(\mathbb{Z})$, the surface going to its transcendental lattice. A selected point is therefore named by its T . Takatsu’s paper on the singular $K3$ surfaces of discriminant 3, 4 and 7 states that each of these discriminants carries exactly one surface, called X_3 , X_4 , X_7 , and tabulates their models. Our checker parses that

table and the uniqueness sentence from the pinned text and, independently, enumerates the reduced forms of each determinant.

Family	z	T (Gram)	$\det T$	classes	Surface
cooper_s7	∞^*	$[2, 1; 1, 2]$	3	1	X_3
cooper_s7	-1	$[2, 1; 1, 4]$	7	1	X_7
cooper_s7	$1/27$	$[2, 0; 0, 14]$	28	2	not named by $\det T$ alone
cooper_s10	∞^*	$[2, 0; 0, 2]$	4	1	X_4

Table 2: Identification of the s_7 points and the s_{10} pick. *: the point picked by the adopted selector. “classes”: the number of reduced even positive-definite forms of that determinant, computed. At $z = 1/27$ two classes share $\det T = 28$, so the determinant alone names no surface there.

Headline result

The s_7 pick is the singular $K3$ surface X_3 ($T = A_2$, $\det T = 3$); the s_{10} pick is X_4 ($T = \langle 2 \rangle \oplus \langle 2 \rangle$, $\det T = 4$). These are the two smallest determinants any even positive-definite binary lattice attains. The s_{10} family cannot reach $\det 3$, because A_2 does not occur in it (a lattice fact recorded in an earlier certificate, not a preference). Tier B — derived, not measured for the identification of T ; Tier L — literature for the bijection and Takatsu’s table.

Remark 3.1. The name “Vinberg’s two most algebraic $K3$ surfaces” for the surfaces of discriminant 3 and 4 is taken from the Schütt–Shioda survey (Tier L — literature). Vinberg’s 1983 paper itself was not fetched or read. The explicit M_7 -polarized model of the 2026-09-29 note gives $E_\omega \times E_\omega$ at $z = \infty$, which matches Takatsu’s construction of X_3 from $E_3 \times E_3$; this match is forced by the Shioda–Inose structure and is reported as consistency, not as independent evidence.

4 What “ $K3 \times T^2$ ” can mean here

Two readings were put to the decision authority on 2026-09-16. Under the literal product reading, the T^2 factor is a fixed torus unrelated to the $K3$, and every $K3$ surface serves equally well, so no selection follows from it. Under the Shioda–Inose reading, which this programme uses, “ T^2 ” is the elliptic side of the pairing the programme has certified: the order-three operator is the symmetric square of an order-two operator, and a member of Picard number 19 is paired with a product $E \times E'$ of two elliptic curves related by a cyclic isogeny of degree n , with $T \cong U \oplus \langle 2n \rangle$. Selection then becomes a question about n , which can be checked.

The check adopted for this reading is unscored: it compares the level n read from the lattice certificate (Gram determinant and a serialized U -splitting witness, both recomputed) with the level read from the modular side (an eta-quotient Hauptmodul for $\Gamma_0(n)^+$ or $\Gamma_0(n)^*$, with Newman’s conditions and an Atkin–Lehner check). Table 3 gives the result. Its power to fail rests on real negative cases: four order-three entries of the same register fail the modular leg.

Family	n from lattice	n from modular side	Verdict
cooper_s7	7	7	T3_AGREE($n=7$)
cooper_s10	10	10	T3_AGREE($n=10$)

Table 3: Lattice level against modular level. An unscored consistency gate, never a ranking.

The physical reading of $K3 \times T^2$ as a compactification is Tier C — blocked physics and is not used. A separately submitted document linking $K3 \times T^2$ to Mathieu moonshine was checked

claim by claim before this note was written: of twelve integer-sequence identifiers it cited, seven were fabricated or attached to the wrong object. Moonshine plays no part here; in any case the $K3$ elliptic genus is the same for every $K3$ surface, so it cannot separate candidates.

5 The twisted-Weierstrass route

A strict-pullback realization of the s_7 family as a Calabi–Yau threefold over the bases tried was excluded earlier (a structural negative, **Tier B — derived, not measured**). The route kept open is a twisted Weierstrass model whose $K3$ fibres carry the M_7 polarization. Three steps have results.

5.1 TW0: a degree question that conflated two quantities

An earlier plan set a fundamental line-bundle degree $\ell = 2$ “via the symmetric-square theorem”. Recomputing from the operator’s Riemann scheme separates two numbers (table 4). The degree of the Weierstrass model’s fundamental line bundle is $\chi(\mathcal{O}_{K3}) = 2$, from Noether’s formula with $\deg \Delta = 24$ (**Tier A**-level arithmetic, standard). The family-level Hodge bundle over the orbifold curve $X_0(7)^+$ has degree $1/3$, and its square has degree $2/3$ (**Tier B — derived, not measured**). The earlier formula reproduced 1 from a Fuchs-relation identity. The rule now reads $\ell := \chi(\mathcal{O}_{K3}) = 2$.

Quantity	Value
orbifold Euler characteristic of $X_0(7)^+$	$-2/3$
$\deg \omega$ (family Hodge bundle, $\mathbb{Q}$ -degree)	$1/3$
$\deg \omega^2$	$2/3$
$\chi(\mathcal{O}_{K3})$ (elliptic-surface reading of ℓ)	2
$\deg \Delta$ of the Weierstrass model	24

Table 4: The two degrees that the earlier plan had merged, from the s_7 operator’s exponents.

5.2 TW1: a necessary screen on threefold bases

The fibres of the generic member contain two E_8 root configurations. A twisted model over a threefold base B_3 needs two divisors carrying the corresponding Tate vanishing orders ($v(f) \geq 4$, $v(g) \geq 5$, $v(\Delta) = 10$) inside the degree budget set by $-4K$, $-6K$ and $-12K$. Table 5 lists the result. On $\mathbb{P}^3$ the budget alone is met, but any two effective divisors meet in a curve, along which the orders reach $(8, 10, 20)$ and the model is not minimal; the screen fails. On $\mathbb{P}^1 \times \mathbb{P}^2$ and on the projective bundles $\mathbb{P}(\mathcal{O} \oplus \mathcal{O}(n))$ over $\mathbb{P}^2$, two disjoint divisors exist and the screen passes. Passing is a necessary condition only: nothing here constructs a model.

Base B_3	Two-divisor degree screen
$\mathbb{P}^3$	FAIL
$\mathbb{P}^1 \times \mathbb{P}^2$	PASS
$\mathbb{P}(\mathcal{O} \oplus \mathcal{O}(n))$ over $\mathbb{P}^2$	PASS (degree budget all $n \geq 0$; realizability checked $n \leq 18$)

Table 5: Two-divisor degree screen. The certificates were written on 2026-07-29 and their status string still reads “DRAFT”; the screen was made live by ruling D20’ on 2026-10-07 (section 7).

5.3 TW2: the section that the M_7 polarization requires

On a $K3$ surface with an elliptic fibration having two E_8 root configurations, the trivial lattice has rank 18 and discriminant ± 1 . Reaching $|\text{disc NS}| = 2n$ with Picard number 19 then needs one

Mordell–Weil section P with height $h(P) = 2n$. The height formula (Schütt–Shioda Thm. 11.5) gives $h(P) = 2\chi + 2\bar{P} \cdot \bar{O} - \sum \text{contr}_v(P)$, so with $\chi = 2$ and no contributions, $\bar{P} \cdot \bar{O} = n - 2$: for s_7 , $h = 14$ and $\bar{P} \cdot \bar{O} = 5$. Kumar–Kuwata give an independent route for $E \not\cong E'$: the section attached to an isogeny of degree d has height $2d$, and $d = 7$ gives the same 14.

At the three $\rho = 20$ points the two curves become isomorphic and the count changes. Exact reduction of the explicit model at each point gives the extra root lattice; the height formula then fixes the section (table 6).

s_7 member	$ \text{disc } T $	ρ	extra roots	MW rank	$h(P)$	$\bar{P} \cdot \bar{O}$
generic	14	19	—	1	14	5
$z = 1/27$	28	20	A_1	1	14	5
$z = -1$	7	20	A_1	1	7/2	0
$z = \infty$	3	20	A_2	0	—	no section

Table 6: The section required by the M_7 polarization: generic member and the three $\rho = 20$ points of s_7 . Root lattices from exact reduction of the explicit model; heights from Schütt–Shioda Thm. 11.5 (Tier B — derived, not measured).

Headline result

The section with $\bar{P} \cdot \bar{O} = 5$ exists on the generic member and at $z = 1/27$. At $z = -1$ the extra A_1 absorbs part of the height and $\bar{P} \cdot \bar{O} = 0$. At $z = \infty$, the point the selector picked, the extra lattice is A_2 , the Mordell–Weil rank is 0 and there is no section at all. The selected surface X_3 therefore does not carry the twisted-route section; the selection and the route are separate questions.

At $z = 1/27$ the elliptic curve has $j = 16581375$ and complex multiplication by the order of discriminant -28 . Using Vélu’s formulas over $\mathbb{Q}$ we wrote an explicit 7-isogeny φ from this curve to a curve with the same j , and checked $\varphi \circ \varphi$ against multiplication by 7 at 60 rational points, more than the degree bound of 49. The x -coordinates fix $\varphi \circ \varphi$ only up to sign, $\pm[7]$. The sign is -7 because the endomorphism algebra is the imaginary quadratic field of discriminant -28 , which has no element of square $+7$ (Tier B — derived, not measured , through the CM certificate). The remaining step, carrying the graph of φ through the Kumar–Kuwata construction down to the $K3$ surface and reading $\bar{P} \cdot \bar{O} = 5$ on the pinned model, is specified but not done.

6 What this note gives the dark-matter stream

Nothing physical. The selected surfaces X_3 and X_4 , the $K3 \times T^2$ reading and the section computation are mathematics about lattices and elliptic surfaces; no CM point and no selected surface maps to any dark-matter observable, and no number here may be used as a prior, a parameter or a fit input. The companion Stream 3 note (repository [DarkMatterK3-Home.github.io](https://github.com/DarkMatterK3-Home), 2026-10-08) reports calibration checks and a literature recount for a mixed fuzzy-dark-matter search window; it uses no result of this note.

7 Corrections recorded in band

Corrections recorded in band

1. The table generator of the 2026-09-29 note attached “(ADVISORY)” to the s_{10} rows by family name. The PDF rebuilt on 2026-10-07 therefore still printed three tags, while its own dated correction said the tables showed none. The generator now reads each row’s `advisory_family` flag (false for every row), a control fails if the tag is keyed on anything else, and that note carries a dated item 7.
2. The same generator typed the discriminant-root orders of its fibration table as literals, against the rule that numbers are computed. The printed values were right; they are now read from the certificate, and a tampered order changes the table (control).
3. On 2026-10-07 a record stated that $\varphi \circ \varphi = [-7]$ was proved. The computation fixed only $\pm[7]$; the sign comes from the CM discriminant. The record carries a dated correction.
4. The author of the discriminant 3, 4, 7 paper was first taken from a search-result snippet, which named other authors; it was corrected from the paper’s own header before any certificate was emitted.
5. The TW1 certificates still read “DRAFT – pending” although ruling D20’ made the screen live. They are left unchanged, as provenance; their status is the one the ledger records.

8 Reproducibility

Every table is generated from `data/certificates/` by `scripts/render_paper_tables.py`, whose `-check` mode fails if any fragment drifts from its certificate; its controls include tampered inputs for every new table. The prose was linted by `scripts/check_paper_tier_language.py`, which applies the programme’s tier-language rules to the whole body and blocks Kodaira fibre-type tokens; it ships a self-test with planted violations. Decision records (D19’–D28’) and the identification brief:

- `briefs/T0_DECISIONS_2026_10_07_STREAM2_DELEGATED.md`
- `briefs/SELECTED_K3_IDENTIFICATION_2026_10_08.md`

Provenance

Generated-by: Claude (Opus 5.5), Stream 2 | Verified-by: `render_paper_tables.py -check` and its controls; each cited checker’s own controls | Reviewed-by: N (preprint; not reviewed).

References

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- [8] X. Callens and the SocrateAI Scientific Agora Collaboration, “Certified lattice data and a within-family selector for the Cooper s_7/s_{10} $K3$ families,” Stream 2 note, 2026-09-29 ([papers/stream2_selection_geometry_2026_09_29.pdf](#)).